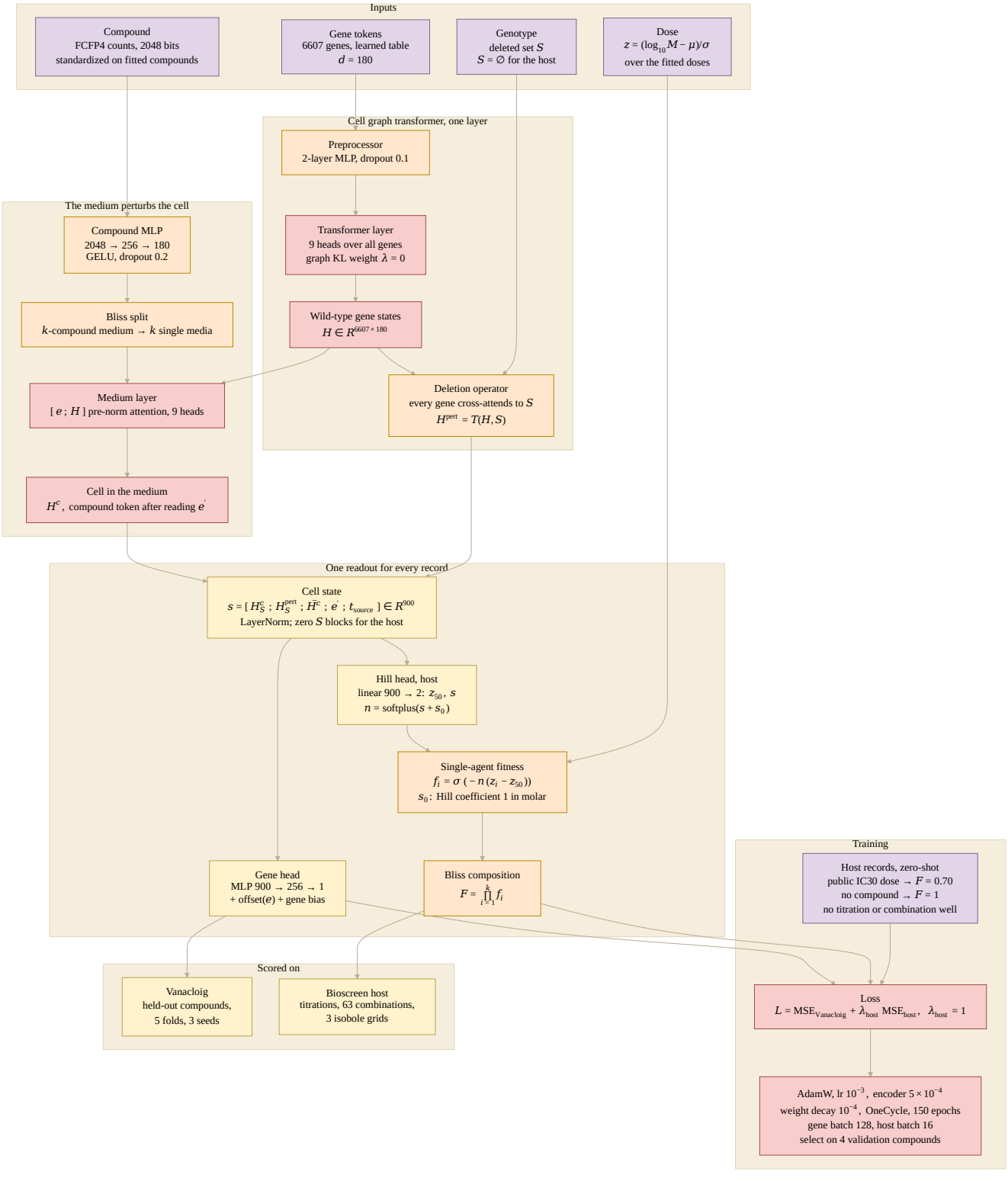

Inputs
Compound
FCFP4 counts, 2048 bits
standardized on fitted compounds
Gene tokens
6607 genes, learned table
d = 180
Genotype
deleted set S
$S = \varnothing$ for the host
Dose
$z = (\log_{10} M - \mu)/\sigma$
over the fitted doses
Cell graph transformer, one layer
Preprocessor
2-layer MLP, dropout 0.1
Transformer layer
9 heads over all genes
graph KL weight λ = 0
Wild-type gene states
$H \in R^{6607 \times 180}$
The medium perturbs the cell
Compound MLP
2048 → 256 → 180
GELU, dropout 0.2
Bliss split
k-compound medium → k single media
Medium layer
[e ; H] pre-norm attention, 9 heads
Cell in the medium
H^c, compound token after reading e'
Deletion operator
every gene cross-attends to S
$H^{pert} = T(H, S)$
One readout for every record
Cell state
$s = [\, H_S^c \,;\, H_S^{pert} \,;\, \bar{H}^c \,;\, e' \,;\, t_{source} \,] \in R^{900}$
LayerNorm; zero S blocks for the host
Hill head, host
linear 900 → 2: z_{50}, s
n = softplus($s + s_0$)
Single-agent fitness
$f_i = \sigma\,(-n\,(z_i - z_{50}))$
s_0: Hill coefficient 1 in molar
Gene head
MLP 900 → 256 → 1
+ offset(e) + gene bias
Bliss composition
$F = \prod_{i=1}^{k} f_i$
Scored on
Vanacloig
held-out compounds,
5 folds, 3 seeds
Bioscreen host
titrations, 63 combinations,
3 isobole grids
Training
Host records, zero-shot
public IC30 dose → F = 0.70
no compound → F = 1
no titration or combination well
Loss
$L = \mathrm{MSE}_{\mathrm{Vanacloig}} + \lambda_{\mathrm{host}}\,\mathrm{MSE}_{\mathrm{host}}$, $\lambda_{\mathrm{host}} = 1$
AdamW, lr 10^{-3}, encoder 5×10^{-4}
weight decay 10^{-4}, OneCycle, 150 epochs
gene batch 128, host batch 16
select on 4 validation compounds